

Unit 8, Lesson 12: Finding Surface Area and Volume

Benchmarks

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.

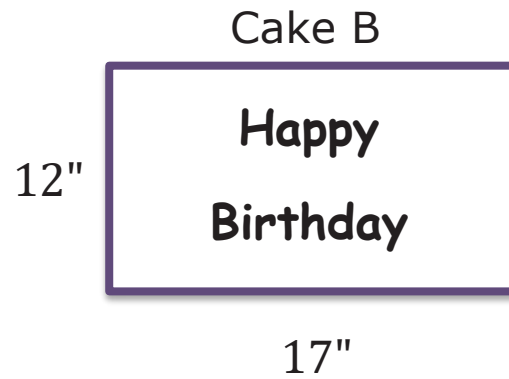
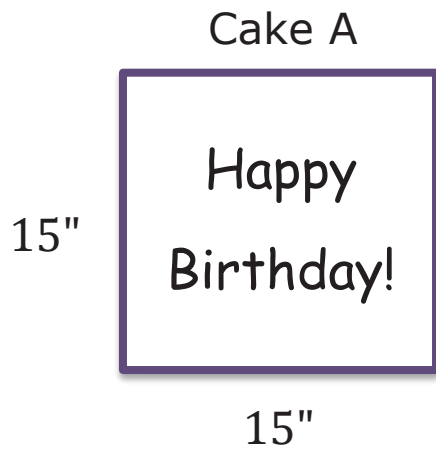
MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.

Learning Targets

- ☐ I can find the surface area of right rectangular prisms and right rectangular pyramids.
- ☐ I can find the volume of right rectangular prisms.

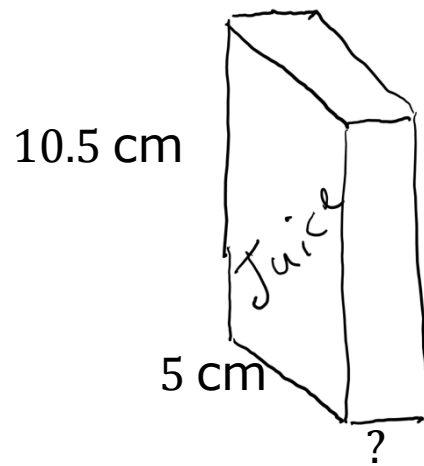
8.12.1 Warm-Up: Would You Rather?

1. Cake A and Cake B are shown below. Cake A is cut into 8 equal slices. Cake B is cut into 6 equal slices. Would you rather have a slice from Cake A or Cake B?



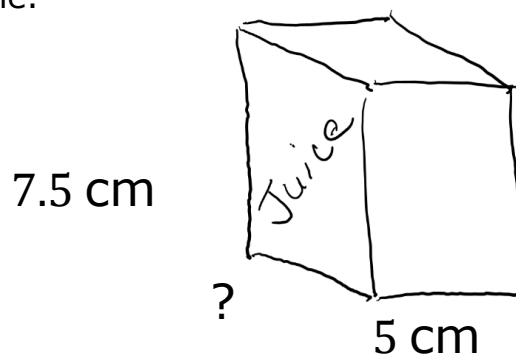
8.12.2 Collaboration: Which Juice Box Is Better?

Julius is designing a juice box. He drew a model of his box.



1. If the juice box has a volume of 196.875 cubic centimeters (cm^3), what is the width of the box?
2. How much packaging board would Julius need to make a juice box with these dimensions, not taking into consideration any overlap of the faces?

Julius wondered how the juice box would be affected if he changed the dimensions, but kept the volume the same.

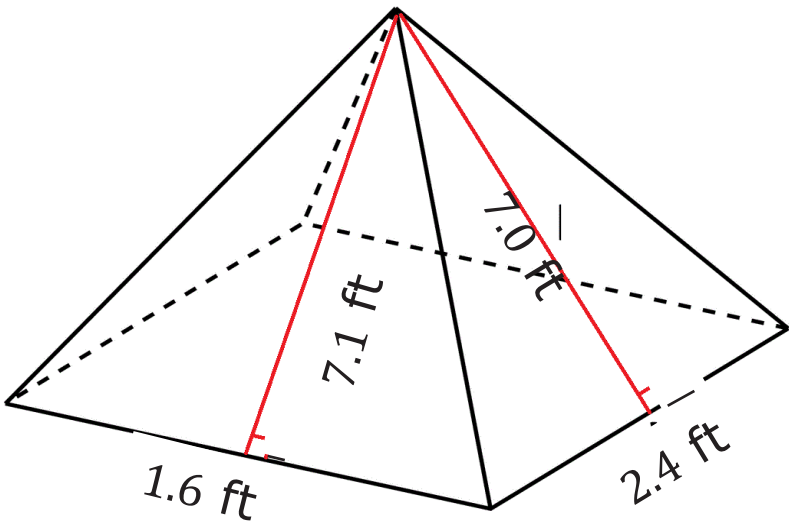


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8.12.3 Guided Practice: Surface Area and Volume

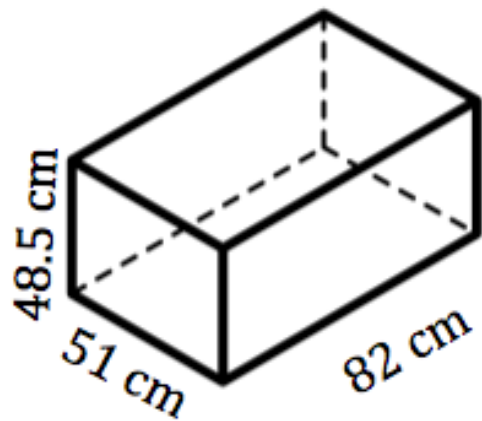
With your partner, complete the chart for the three-dimensional figures below.

1.



What is the type of figure?	
Draw a net or the individual faces of the figure. Label the measurements.	
Find the surface area of the figure.	

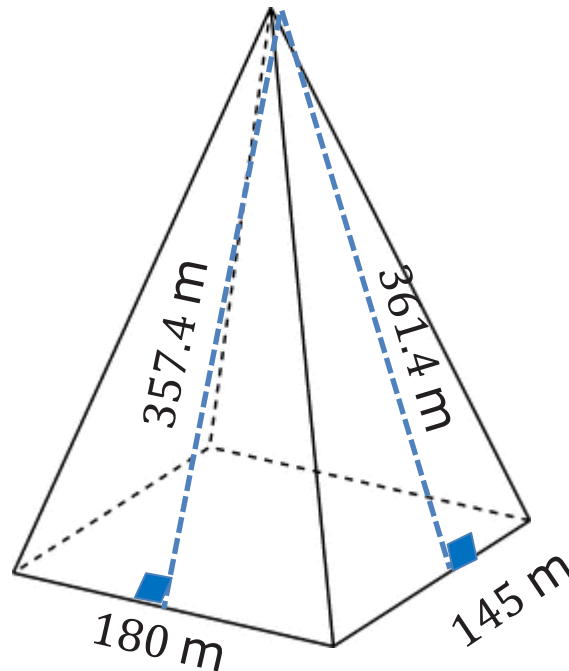
2.



What is the type of figure?	
Draw a net of the figure or the individual faces. Label the measurements.	
Find the surface area of the figure.	
Find the volume of the figure.	

8.12.4 Your Turn!

1. Nikolas is constructing a skyscraper with walls made of only glass that is 350 meters (m) high and is in the shape of a rectangular pyramid.

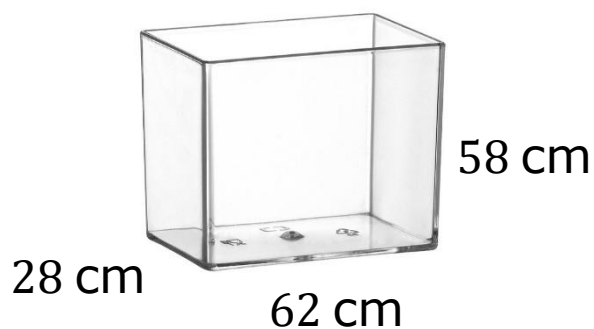


- a. What is the surface area of the skyscraper?
 - b. Using the model shown, determine how much of the surface area will be made of glass.
2. A cooler has a length of 78 centimeters (cm), a height of 74 cm, and a width of 40 cm.

3.



- Find the surface area of the cooler.
- Find the volume of the cooler.
- If the **inside** of the cooler has a length 62 cm, a height of 58 cm, and a width of 28 cm, how much smaller is the volume of the inside of the cooler than the volume of the entire cooler?



- Tamya is creating a gardening box that needs to have a volume of 13.5 ft^3 .

- a. If the box has a width of 3 feet and a length of 2 feet, what is the height?



- b. Tanya decided to paint the exterior walls of the gardening box and needs to know how much paint to buy. What is the surface area of the area she wants to paint?

5. Ryan makes custom jewelry and recently received an order for a pendant in the shape of a rectangular pyramid. Ryan designed a hollow gold pendant to have a square base with a side length of 15 millimeters (mm) and a slant height of 13.24 mm.



How much gold does Ryan need to make the pendant?

8.12.5 Wrap-Up: Reflection

1. Rate yourself on each of the concepts.

- 4 – I can do this on my own.
- 3 – I can do this but need more practice.
- 2 – I need help getting started.
- 1 – I do not understand yet.

Concept	Rating
I can find the surface area of a right rectangular pyramid using a net or the figure.	1234
I can find the surface area of a right rectangular prism using a net or the figure.	1234
I can find the volume of a right rectangular prism.	1234

2. A barrier to my learning today was ...

3. I will try to overcome this by ...